

Deep Learning-Based Athlete Injury Risk Analyzer

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Abstract

This work presents a deep-learning system that estimates athlete injury risk from 15 biometric, training, and lifestyle variables. A Multi-Layer Perceptron (MLP), regularized with batch normalization and dropout, is trained on 1,000 athlete records. The system adds SHAP and LIME explanations, personalized recommendations, and a Streamlit interface. On the held-out test set, the DNN achieves 0.8850 accuracy and 0.9108 ROC-AUC, exceeding a Logistic Regression baseline.

Keywords— *deep learning, MLP, injury prediction, explainable AI, SHAP, LIME, Streamlit.*

1. Introduction

Sports injuries affect athlete well-being and performance. Conventional assessment can be subjective and inaccessible. This project provides a reproducible, data-driven risk classifier that uses readily available athlete information and explains individual predictions for coaches and athletes.

A. Objectives

The objectives are to predict high injury risk, evaluate the model against a conventional baseline, explain each prediction, and provide a deployable interface with practical recommendations.

2. Related Work

Prior work applies machine learning to injury prediction using GPS, wearable, and training-load data [1]–[6]. The literature shows useful predictive performance, but commonly lacks a simple tabular-data deployment, DNN explanations, or personalized recommendations. The present work combines an MLP, post-hoc SHAP/LIME explanations, and a Streamlit application.

Study	Method	Gap addressed here
Rossi et al. [1]	ML with GPS data	No DNN/XAI
Claudino et al. [2]	Review of AI in sports	Need accessible deployment
Kim and Park [5]	MLP/LSTM with IMU data	Requires wearables; no XAI
Torres et al. [6]	DNN and XGBoost	No real-time XAI application

3. Methodology

A. Dataset and Preprocessing

The supplied *Athlete.xlsx* dataset contains 1,000 records, 15 input features, and a binary Injury_Risk target. It includes age, anthropometrics, training frequency and intensity, sleep, flexibility, muscle asymmetry, recovery, injury history, and stress. Categorical fields are label

encoded; numeric features are standardized. A stratified 80/20 split preserves the 80.9% low-risk and 19.1% high-risk class distribution.

B. Architecture and Module Mapping

Module 1: a 4-hidden-layer MLP. **Module 2:** a trained feature-reconstruction autoencoder (15→64→32→16→32→64→15) with a 16-unit latent representation. **Module 3:** SHAP/LIME explanations, personalized recommendations, and Streamlit deployment.

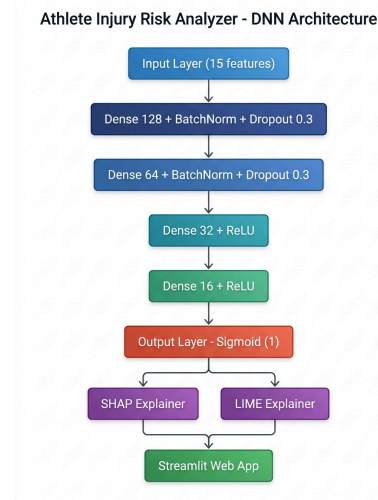


Fig. 1. MLP architecture: 15 standardized inputs, Dense(128)/BatchNorm/Dropout, Dense(64)/BatchNorm/Dropout, Dense(32), Dense(16), and sigmoid output.

C. Training

The model uses Adam (learning rate 0.001), binary cross-entropy, early stopping, learning-rate reduction, and checkpointing. The selected checkpoint is stored as *models/best_model.keras*.

4. Results and Discussion

The held-out test set contains 200 samples. Table I compares the trained DNN to Logistic Regression. The DNN improves every reported metric, particularly precision and F1 for the minority high-risk class.

TABLE I. HELD-OUT TEST RESULTS

Metric	DNN	Logistic Regression	Improvement
Accuracy	0.8850	0.8600	0.0250
Precision	0.6923	0.6250	0.0673
Recall	0.7105	0.6579	0.0526
F1 score	0.7013	0.6410	0.0603
ROC-AUC	0.9108	0.8970	0.0138

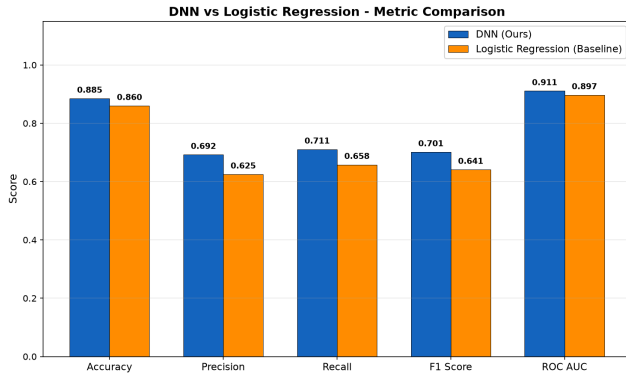


Fig. 2. The DNN outperforms the Logistic Regression baseline in all metrics.

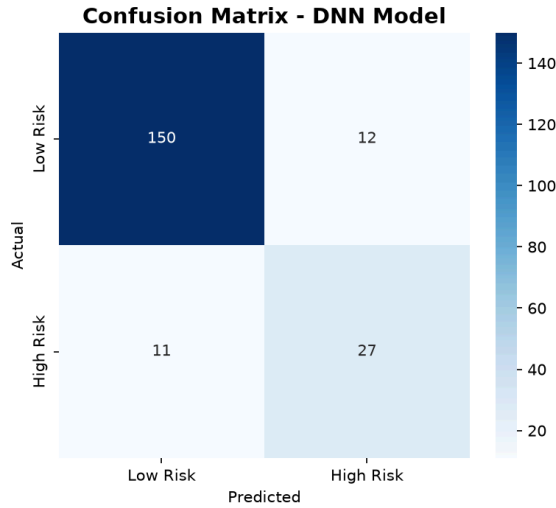


Fig. 3. Confusion matrix for the DNN at a 0.5 decision threshold.

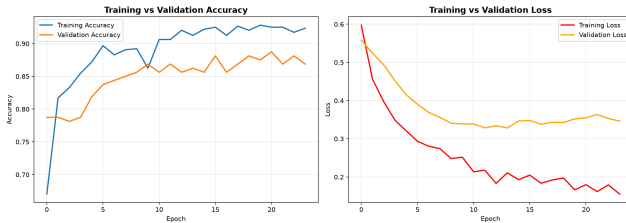


Fig. 4. Training and validation accuracy/loss curves from the evaluated training run.

A. Real-World Considerations

The high-risk class is underrepresented; therefore F1, recall, and AUC complement accuracy. The self-reported features may introduce measurement error, and the data may not represent every sport, region, age group, or competition level. Future work should collect larger multi-sport cohorts, evaluate calibration, use class weighting/SMOTE where appropriate, and validate prospectively.

5. Conclusion

The project demonstrates an end-to-end, explainable injury-risk workflow. The DNN achieved 0.8850 accuracy and 0.9108 ROC-AUC, exceeding Logistic Regression on the held-out test set. The Streamlit interface and SHAP/LIME explanations make predictions accessible, while the stated data limitations define clear next steps for responsible real-world use.

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